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Adjustment arrangement for offset spacing between louvres in a sieve of an agricultural harvester

Abstract

A sieve for use in an agricultural harvester includes first and second adjustment laths arranged longitudinally adjacent to each other. The first and second adjustment laths are movable in fore and aft directions relative to an air flow direction (Y) through the sieve. The first adjustment lath is connected with a first set of louvres and the second adjustment lath is connected with a second set of louvres. An adjustment arrangement for adjusting an offset spacing (X) between first and second sets of louvres includes a rotatable elongate element with a longitudinally facing stepped shoulder and a stop surface. The elongate element is connected with each of the first and second adjustment laths. The stepped shoulder includes a plurality of stepped surfaces which are selectively engagable with the at least one stop surface. The stepped surfaces respectively define incremental offset spacings (X) between the first and second set of louvres.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to European Patent Application No. 21175441.1, filed May 21, 2021, the content of such application being incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present disclosure relates to a cleaning system used in an agricultural harvester, such as a combine, and, more particularly, to a sieve used in such a cleaning system.

2. Description of the Related Art

(3) An agricultural harvester combines multiple harvesting functions with a single harvesting unit, such as picking, threshing, separating and cleaning. A combine includes a header which removes the crop from a field, and a feeder housing which transports the crop matter into a threshing rotor. The threshing rotor rotates within a perforated housing, which may be in the form of adjustable concaves, and performs a threshing operation on the crop to remove the grain. Once the grain is threshed it falls through perforations in the concaves onto a grain pan. From the grain pan the grain is cleaned using a cleaning system, and is then transported to a grain tank onboard the combine. The cleaning system includes a cleaning fan which blows air through oscillating sieves to discharge chaff and other debris toward the rear of the combine. Non-grain crop material such as straw from the threshing section proceeds through a straw chopper and out the rear of the combine. When the grain tank becomes full, the combine is positioned adjacent a vehicle into which the grain is to be unloaded, such as a semi-trailer, gravity box, straight truck, or the like; and an unloading system on the combine is actuated to transfer the grain into the vehicle.

(4) More particularly, a rotary threshing or separating system includes one or more rotors which can extend axially (front to rear) or transversely within the body of the combine, and which are partially or fully surrounded by a perforated concave. The crop material is threshed and separated by the rotation of the rotor within the concave. Coarser non-grain crop material such as stalks and leaves are transported to the rear of the combine and discharged back to the field. The separated grain, together with some finer non-grain crop material such as chaff, dust, straw, and other crop residue are discharged through the concaves and fall onto the grain pan where they are transported to the cleaning system. Alternatively, the grain and finer non-grain crop material may also fall directly onto the cleaning system itself.

(5) The cleaning system further separates the grain from non-grain crop material, and typically includes a fan directing an air flow stream upwardly and rearwardly through vertically arranged sieves which oscillate in a fore and aft manner. The air flow stream lifts and carries the lighter non-grain crop material towards the rear end of the combine for discharge to the field. Clean grain, being heavier, and larger pieces of non-grain crop material, which are not carried away by the air flow stream, fall onto a surface of an upper sieve (also known as a chaffer sieve) where some or all of the clean grain passes through to a lower sieve (also known as a cleaning sieve). Grain and non-grain crop material remaining on the upper and lower sieves are physically separated by the reciprocating action of the sieves as the material moves rearwardly. Any grain and/or non-grain crop material remaining on the top surface of the upper sieve are discharged at the rear of the combine. Grain falling through the lower sieve lands on a bottom pan of the cleaning system, where it is conveyed forwardly toward a clean grain auger.

(6) The clean grain auger is positioned below the lower sieve, and receives clean grain from each sieve and from the bottom pan of the cleaning system. The clean grain auger then augers the clean grain laterally sideways to a clean grain elevator, which in turn conveys the clean grain to a grain tank onboard the combine.

(7) A sieve typically includes a plurality of louvres extending cross-wise between the sides of the sieve, with the louvres including a plurality of fingers which are attached at their rear edge to respective wires. The louvres can be rotated, which in turn opens and closes the spacing between the adjacent rows of fingers. Depending on the crop being harvested, size of the kernels, amount of non-crop matter, etc, the louvre spacing (between the fingers) may need to be adjusted during operation. To that end, the louvres can be connected to at least one adjustment lath which can be moved forward and rearward. The adjustment lath can be connected by way of hinges, levers, etc. to a manual adjustment mechanism, such as a lever at the rear of the sieve which allows an operator

to manually adjust the louvre spacing.

(8) It is also known to separate the louvres on sieves into discrete sections of louvers which can be independently adjusted. For example, the upper sieve can be separated into main louvres toward the upstream end of the sieve and extension louvres toward the downstream end of the sieve or vice versa. The extension louvres can be made up, for example, of 10 rows of louvres which can be adjusted to adjust the spacing between the louvres. Likewise, the main louvres consisting of the remaining rows of louvres can be separately adjusted to adjust the spacing between the louvres. The adjustment of both the main louvres and the extension louvres are independent from each other.

SUMMARY OF THE INVENTION

(9) The present disclosure provides an adjustment arrangement for a sieve which simultaneously adjusts an offset spacing between main sieve louvres and extension louvres.

(10) The disclosure in one form is directed to a sieve for use in an agricultural harvester. The sieve includes a framework with a pair of opposed side members. A plurality of louvres extend between the pair of opposed side members. The louvres are arranged in a first set of louvres and a second set of louvres, with each of the first set of louvres and the second set of louvres including a plurality of adjacent louvres overlapped with each other. The second set of louvres are disposed downstream from the first set of louvres relative to an air flow direction through the sieve during operation. The sieve is characterized by a first adjustment lath and a second adjustment lath arranged longitudinally adjacent to each other. The first adjustment lath and the second adjustment lath are movable in fore and aft directions relative to the air flow direction through the sieve. The first adjustment lath is connected with the first set of louvres and the second adjustment lath is connected with the second set of louvres. An adjustment arrangement for adjusting an offset spacing between the first set of louvres and the second set of louvres includes a rotatable elongate element with a longitudinally facing stepped shoulder and at least one stop surface. The elongate element is connected with each of the first adjustment lath and the second adjustment lath. The stepped shoulder includes a plurality of stepped surfaces which are selectively engagable with the at least one stop surface. The stepped surfaces respectively define incremental offset spacings between the first set of louvres and the second set of louvres.

(11) In another embodiment, the elongate element comprises a rod, and a first bushing is carried by the rod and defines the stepped shoulder.

(12) In another embodiment, a second bushing is carried by the rod which interfaces with the first bushing and defines the at least one stop surface. The at least one stop surface includes a plurality of mating stepped surfaces which interface with the stepped surfaces on the first bushing.

(13) In another embodiment, the first bushing includes ramped surfaces between the stepped surfaces, and the second bushing includes ramped surfaces between the mating stepped surfaces.

(14) In another embodiment, the first bushing is mounted to the rod, and the second bushing is mounted to the second adjustment lath.

(15) In yet another embodiment, the adjustment arrangement further includes a spring positioned around the rod and extending between and engaging each of the first adjustment lath and the second adjustment lath, thereby biasing the first adjustment lath and the second adjustment lath away from each other.

(16) In yet another embodiment, the first adjustment lath and the second adjustment lath each include a tube and the rod extends through the tubes, and the spring is positioned between and engages the tubes.

(17) In a further embodiment, the spring is a compression spring.

(18) In a further embodiment, the sieve comprises an upper sieve.

(19) In a still further embodiment, the first set of louvres comprise main sieve louvres and the second set of louvres comprise extension sieve louvres.

(20) In a still further embodiment, a method of operating a sieve in an agricultural harvester is carried out using the adjustment arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 is a side view of an embodiment of an agricultural harvester in the form of a combine which may include a cleaning system of the present disclosure;

(3) FIG. 2 is a perspective view of an embodiment of a sieve of the present disclosure which can be used in the harvester of FIG. 1;

(4) FIG. 3 is a bottom view showing an embodiment of an adjustment arrangement of the present disclosure;

(5) FIG. 4 is a side view of the adjustment arrangement shown in FIG. 3;

(6) FIG. 5 is a side view of the adjustment arrangement shown in FIGS. 3 and 4, with the offset spacing the same between the main louvres and the extension louvres;

(7) FIG. 6 is a side view of the adjustment arrangement shown in FIGS. 3-5, with a maximum offset spacing between the main louvres and the extension louvres; and

(8) FIG. 7 is a perspective view of a bushing forming part of the adjustment arrangement shown in FIGS. 3-6.

(9) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate embodiments of the invention, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE INVENTION

(10) The terms “grain”, “straw” and “tailings” are used principally throughout this specification for convenience but it is to be understood that these terms are not intended to be limiting. Thus “grain” refers to that part of the crop material which is threshed and separated from the discardable part of the crop material, which is referred to as non-grain crop material, MOG or straw. Incompletely threshed crop material is referred to as “tailings”. Also the terms “forward”, “rearward”, “left” and “right”, when used in connection with the agricultural harvester and/or components thereof are usually determined with reference to the direction of forward operative travel of the harvester, but again, they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the agricultural harvester and are equally not to be construed as limiting.

(11) Referring now to the drawings, and more particularly to FIG. 1, there is shown an agricultural harvester in the form of a combine **10**, which generally includes a chassis **12**, ground engaging wheels **14** and **16**, header **18**, feeder housing **20**, operator cab **22**, threshing and separating system **24**, cleaning system **26**, grain tank **28**, and unloading auger **30**.

(12) Front wheels **14** are larger flotation type wheels, and rear wheels **16** are smaller steerable wheels. Motive force is selectively applied to front wheels **14** through a power plant in the form of a diesel engine **32** and a transmission (not shown). Although combine **10** is shown as including wheels, is also to be understood that combine **10** may include tracks, such as full tracks or half tracks.

(13) Header **18** is mounted to the front of combine **10** and includes a cutter bar **34** for severing crops from a field during forward motion of combine **10**. A rotatable reel **36** feeds the crop into header **18**, and a double auger **38** feeds the severed crop laterally inwardly from each side toward

feeder housing **20**. Feeder housing **20** conveys the cut crop to threshing and separating system **24**, and is selectively vertically movable using appropriate actuators, such as hydraulic cylinders (not shown).

(14) Threshing and separating system **24** is of the axial-flow type, and generally includes a rotor **40** at least partially enclosed by and rotatable within a corresponding perforated concave **42**. The cut crops are threshed and separated by the rotation of rotor **40** within concave **42**, and larger elements, such as stalks, leaves and the like are discharged from the rear of combine **10**. Smaller elements of crop material including grain and non-grain crop material, including particles lighter than grain, such as chaff, dust and straw, are discharged through perforations of concave **42**. Threshing and separating system **24** can also be a different type of system, such as a system with a transverse rotor rather than an axial rotor, etc.

(15) Grain which has been separated by the threshing and separating assembly **24** falls onto a grain pan **44** and is conveyed toward cleaning system **26**. Cleaning system **26** may include an optional pre-cleaning sieve **46**, an upper sieve **48** (also known as a chaffer sieve), a lower sieve **50** (also known as a cleaning sieve), and a cleaning fan **52**. The upper sieve **48** and lower sieve **50** can be carried within a common framework, and are generally designated together as sieve assembly **100**, which will be described in more detail hereinafter. Grain on sieves **46**, **48** and **50** is subjected to a cleaning action by fan **52** which provides an air flow through the sieves to remove chaff and other impurities such as dust from the grain by making this material airborne for discharge from straw hood **54** of combine **10**. Grain pan **44** and pre-cleaning sieve **46** oscillate in a fore-to-aft manner to transport the grain and finer non-grain crop material to the upper surface of upper sieve **48**. Upper sieve **48** and lower sieve **50** are vertically arranged relative to each other, and likewise oscillate in a fore-to-aft manner to spread the grain across sieves **48**, **50**, while permitting the passage of cleaned grain by gravity through the openings of sieves **48**, **50**.

(16) Clean grain falls to a clean grain auger **56** positioned crosswise below and toward the front of lower sieve **50**. Clean grain auger **56** receives clean grain from each sieve **48**, **50** and from bottom pan **58** of cleaning system **26**. Clean grain auger **56** conveys the clean grain laterally to a generally vertically arranged grain elevator **60** for transport to grain tank **28**. Tailings from cleaning system **26** fall to a tailings auger trough **62**. The tailings are transported via tailings auger **64** and return auger **66** to the upstream end of cleaning system **26** for repeated cleaning action. A pair of grain tank augers **68** at the bottom of grain tank **28** convey the clean grain laterally within grain tank **28** to unloading auger **30** for discharge from combine **10**.

(17) Referring now to FIGS. 2 and 3, the sieve assembly **100** includes a cleaning shoe **101** in which the upper sieve **48** and lower sieve **50** can be removably installed. The upper sieve **48** and lower sieve **50** each generally include a framework **102** with a pair of opposed side members **104** (only one of which is visible for each sieve **48**, **50** in FIG. 2). A first cross member **108** extends between the side members **104** at one end of the side members **104** and a second cross member **110** extends between the side members **104** at an opposite end of the side members **104**. In the embodiment shown, the first cross member **108** is positioned at the downstream end of the sieve **48** or **50**, and the second cross member **110** is positioned at the upstream end of the sieve **48** or **50**.

(18) The upper sieve **48** and the lower sieve **50** each include a plurality of louvres **112** (FIGS. 2 and 3; sometimes also referred to as wires) which extend between the pair of opposed side members **104**. The louvres **112** are pivotally mounted at opposite ends to the side members **104**. Each louver **112** includes a plurality of fingers **114** which overlap relative to and can be offset from the fingers **114** of adjacent louvres. The spacing between the fingers **114** of adjacent louvres can be adjusted by simultaneously rotating the louvres within the side members **104**.

(19) For some applications, it may be desirable to separate or subdivide the louvres of a sieve into discrete sections, relative to the air flow direction Y through the upper sieve. For example, it may be desirable to adjust the louver spacing differently for different parts of a sieve, based on grain and MOG size, weight, etc. Referring now to FIGS. 3-6, the upper sieve **48** can be separated into two or

more sections or groups of louvers **112A**, **112B** relative to the air flow direction Y through the upper sieve **48**. The extension louvers **112B** are assumed to be defined by the last 10 rows of louvers, at the downstream end of the upper sieve **48**. The extension louvers **112B** could be defined by a different number of louvers, such as the last 8 or 11 rows of louvers. The main louvers **112A** are defined by the remaining louvers of the upper sieve **48**, and are located upstream from the extension louvers **112B**. In an alternative embodiment the main louvers **112A** can be positioned downstream of the extension louvers **112B**. The spacing between the main louvers **112A** and the extension louvers **112B** are adjusted dependent on each other, and have an offset spacing X, as will be described in more detail hereinafter. For purposes of discussion herein, the main louvers **112A** are assumed to define a first set of louvers, and the extension louvers **112B** are assumed to define a second set of louvers.

(20) Referring now to FIGS. **4-6**, a first adjustment lath **116A** and a second adjustment lath **116B** are arranged longitudinally adjacent to each other, with the first adjustment lath **116A** and the second adjustment lath **116B** being movable in fore and aft directions relative to the air flow direction Y through the sieve **48**. The first adjustment lath **116A** is connected with the first set of louvers **112A** and the second adjustment lath **116B** is connected with the second set of louvers **112B**.

(21) An adjustment arrangement **120** for adjusting the offset spacing X between the first set of louvers **112A** and the second set of louvers **112B** is positioned below the upper sieve **48**. The adjustment arrangement **120** includes a rotatable elongate element **122** with a longitudinally facing stepped shoulder **124** and at least one stop surface **126**. The elongate element **122** can be in the form of a rod which is connected with each of the first adjustment lath **116A** and the second adjustment lath **116B**. In the illustrated embodiment, the first adjustment lath **116A** includes a single tube **128A** through which the rod **122** extends. Likewise, the second adjustment lath **116B** includes two tubes **128B** through which the rod **122** extends.

(22) A first bushing **130** (FIGS. **3-7**) carried by the rod **122** defines the stepped shoulder **124**. The first bushing **130** can be immovably fixed to the rod **122**, such as by pinning, welding, etc. The stepped shoulder **124** includes a plurality of stepped surfaces **124A** (FIG. **7**) which are selectively engagable with the at least one stop surface **126**. The stepped surfaces **124A** respectively define incremental offset spacings between the first set of louvers **112A** and the second set of louvers **112B**. The first bushing **130** also includes ramped surfaces **124B** between the stepped surfaces **124A**. The ramped surfaces **124B** allow the first bushing **30** to be rotated to a selected position such that a selected stepped surface **124A** engages with the one or more stop surfaces **126**.

(23) A second bushing **132** is also carried by the rod **122**. The second bushing **132** interfaces with the first bushing **130** and defines the at least one stop surface **126**. In the embodiment shown, the second bushing **132** is slidable on the rod **122** and has a notched end which engages with a projection on the tube **128B** of the second adjustment lath **116B**, thereby making the second bushing **132** non-rotatable relative to the second adjustment lath **116B**. The at least one stop surface **126** defined by the second bushing **132** includes a plurality of mating stepped surfaces **126A** which interface with the stepped surfaces **124A** on the first bushing **130**. The second bushing **132** also includes ramped surfaces (not numbered) between the mating stepped surfaces **126A**.

(24) The adjustment arrangement **120** further includes a spring **134** positioned around the rod **122**. The spring **134** extends between and engages each of the first adjustment lath **116A** and the second adjustment lath **116B**, thereby biasing the first adjustment lath **116A** and the second adjustment lath **116B** away from each other. In the illustrated embodiment, the spring **134** is configured as a compression spring which is positioned between and engages the tubes **128A** and **128B**.

(25) In the embodiment of the adjustment arrangement **120** shown and described above, the adjustment arrangement is used in conjunction with the upper sieve **48**. However, it is to be understood that the adjustment arrangement **120** could be used with a different sieve where it is desired to provide an offset spacing between sections of louvers. For example, it may be possible to

use the adjustment arrangement **120** with the pre-sieve **46** or the lower sieve **50**.

(26) Furthermore, in the embodiment of the adjustment arrangement **120** shown and described above, the second bushing **132** defines mating stepped surfaces which interface with the stepped surfaces of the first bushing **130**. However, it may also be possible to eliminate the second bushing **132** and form the mating stepped surfaces as part of the axial face of the adjoining tube **128B**.

(27) During operation, the adjustment arrangement **120** can be manually adjusted by rotating the handle **136** attached to the end of the rod **122**. Alternatively, rod **122** can be rotated by an electric or hydraulic motor instead of handle **136** and can be remotely controlled. Referring to FIG. 5, the offset spacing **X** can be adjusted such that the spacing between the main louvres **112A** is the same as the spacing between the extension louvres **112B** (i.e., the fingers **114** of the extension louvres **112B** are at the same angular orientation as the fingers **114** of the main louvres **112A**). The offset spacing **X** can be selectively adjusted anywhere up to a maximum offset spacing, shown in FIG. 6.

(28) Whenever the first adjustment lath **116** is moved towards its rearmost position rotating the main louvres **112A** towards an extreme position, at some point in the movement the maximum offset spacing **X** is not possible anymore. Compression of the spring **134** covers the reduction of this offset spacing **X**.

Claims

1. A sieve for use in an agricultural harvester, the sieve comprising: a framework including a pair of opposed side members; a plurality of louvres extending between the pair of opposed side members, the louvres being arranged in a first set of louvres and a second set of louvres, each of the first set of louvres and the second set of louvres including a plurality of adjacent louvres overlapped with each other, the second set of louvres being disposed downstream or upstream from the first set of louvres relative to an air flow direction (**Y**) through the sieve during operation; a first adjustment lath and a second adjustment lath arranged longitudinally adjacent to each other, each of the first adjustment lath and the second adjustment lath being movable in fore and aft directions relative to the air flow direction (**Y**) through the sieve, the first adjustment lath being connected with the first set of louvres such that the first set of louvres is carried on the first adjustment lath and the second adjustment lath being connected with the second set of louvres such that the second set of louvres is carried on the second adjustment lath; and an adjustment arrangement for adjusting an offset spacing (**X**) between the first adjustment lath and the second adjustment lath, the adjustment arrangement including a rotatable elongate element with a longitudinally facing stepped shoulder and at least one stop surface, the elongate element being connected with each of the first adjustment lath and the second adjustment lath such that rotation of the elongate element adjusts the offset spacing (**X**) between the first adjustment lath and the second adjustment lath, the stepped shoulder including a plurality of stepped surfaces which are selectively engageable with the at least one stop surface, the stepped surfaces respectively defining incremental offset spacings (**X**) between the first adjustment lath and the second adjustment lath.

2. The sieve of claim 1, wherein the elongate element comprises a rod, and further including a first bushing carried by the rod and defining the stepped shoulder having the plurality of stepped surfaces.

3. The sieve of claim 2, further including a second bushing carried by the rod which interfaces with the first bushing and defines the at least one stop surface, the at least one stop surface includes a plurality of mating stepped surfaces which interface with the stepped surfaces on the first bushing.

4. The sieve of claim 3, wherein the first bushing includes ramped surfaces between the stepped surfaces.

5. The sieve of claim 4, wherein the first bushing is mounted to the rod, and the second bushing is mounted to the second adjustment lath.

6. The sieve of claim 2, wherein the adjustment arrangement further includes a spring extending

between and engaging each of the first adjustment lath and the second adjustment lath, thereby biasing the first adjustment lath and the second adjustment lath away from each other.

7. The sieve of claim 6, wherein the first adjustment lath and the second adjustment lath each include a tube and the rod extends through the tubes, and the spring is positioned between and engages the tubes.

8. The sieve of claim 6, wherein the spring is a compression spring.

9. The sieve of claim 1, wherein the sieve is an upper sieve.

10. The sieve of claim 1, wherein the first set of louvres comprise main sieve louvres and the second set of louvres comprise extension sieve louvres.

11. An agricultural harvester comprising the sieve of claim 1.
